

Crossing the Quantum Boundary: Post-Quantum Custody Continuity with Bounded Classical Delegation

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Research manuscript – full version

Prototype measurements pending; no fabricated performance claims

Abstract—A blockchain may protect its native asset with a post-quantum (PQ) signature and nevertheless lose that protection when the asset is deposited into an Ethereum-compatible execution domain controlled by a classical ECDSA account. We call this failure a *cross-domain cryptographic downgrade*. Existing work addresses atomic cross-ledger transfer, programmable accounts, efficient PQ verification, or replacement of ECDSA, but does not define an end-to-end custody property for an asset that moves between heterogeneous authorization domains while retaining constrained compatibility with classical wallets.

We introduce *post-quantum custody continuity* (PQCC), a game-based property for hybrid UTXO–EVM systems. We then present PREA, a PQ-rooted EVM account construction combining: (i) exclusive UTXO–EVM value representation, (ii) an irreversible PQ account registry and consensus-enforced closure of residual direct-classical authorization paths, and (iii) policy-complete classical session grants whose compromise causes at most an explicitly committed, stateful loss vector. Our adversary is intentionally strong: it may forge every classical signature immediately, control relayers, and exploit any operation admitted by a session policy; only the ML-DSA root, collision-resistant hashing, and finalized ledger safety remain trusted.

We formalize supply conservation, claim/refund exclusion, authentication monotonicity, root authorization, and bounded classical delegation. Under stated assumptions, we prove that an adversary cannot obtain irreversible custody beyond the sum of authorized session caps without either forging ML-DSA, violating ledger safety, finding a hash collision, or exploiting an incomplete policy monitor. We also prove necessity results: mirrored balances violate conservation; any unrestricted classical destination path defeats PQCC; and approval-like side effects invalidate scalar spending caps. A bounded executable model explores 201 strict-design states without violation and automatically produces minimal counterexamples for four weakened variants. Finally, we map the construction to a Rust UTXO–EVM implementation and define the test, benchmark, fuzzing, and artifact gates required for a systems-security submission. This manuscript deliberately reports no unmeasured performance numbers.

Index Terms—post-quantum cryptography, account abstraction, blockchain bridges, hybrid ledgers, session keys, capability security, ML-DSA, EVM

I. INTRODUCTION

Post-quantum migration is often framed as replacing one transaction signature with another. That framing is incomplete for systems that expose more than one execution domain. Consider a chain whose native UTXO ledger authorizes spends with ML-DSA, but whose optional EVM lane uses ordinary Ethereum accounts. A coin can begin under a PQ key, cross

an atomic deposit interface, and emerge under secp256k1. The bridge can conserve supply perfectly while silently degrading the asset’s authorization strength. A sufficiently capable quantum adversary need not break the bridge or consensus: it attacks the newly exposed classical account.

This paper studies that boundary. We ask:

Can a hybrid UTXO–EVM chain preserve post-quantum custody of an asset across an exclusive value transfer, while retaining practical classical session keys for frequent EVM interactions?

The answer is not obtained by merely inserting a PQ verification precompile. Account abstraction already permits alternative validation logic [1], modular account standards already discuss session keys and spending limits [2], and recent Ethereum proposals explore irreversible ECDSA deactivation or native alternative keys [3], [4], [5]. Efficient PQ verification on Ethereum is itself an active research topic [6]. The missing object is an end-to-end security definition tying *source custody*, *exclusive value movement*, *destination authority*, and *bounded compatibility delegation* into one composable statement.

A. Cross-Domain Cryptographic Downgrade

Let authorization schemes form a security preorder. For the threat model of this paper, an ML-DSA root is uncompromised while secp256k1 signatures are fully forgeable. An asset transition

$$(\text{UTXO}, \text{MLDSA}) \longrightarrow (\text{EVM}, \text{ECDSA})$$

therefore decreases the minimum authorization strength needed to steal the asset. Atomicity and supply conservation do not prevent this downgrade. We call the phenomenon *cross-domain cryptographic downgrade* (CDCD).

CDCD differs from a conventional bridge failure. No message needs to be forged, no lock needs to be claimed twice, and no consensus safety property needs to fail. The destination account simply exposes a weaker authorization path than the source. This is especially consequential for assets that remain in the EVM for long periods: NFTs, contract-admin roles, token allowances, and marketplace positions cannot be protected by an “auto-withdraw soon” operational guideline.

B. Our Approach

PREA treats the native PQ UTXO ledger as the long-term vault and the EVM as a programmable work domain. Value moves exclusively: a unit is either a spendable UTXO, an unclaimed deposit lock, an EVM balance, or burned. The EVM destination is a deterministic smart account rooted in an ML-DSA public key. Classical secp256k1 keys are not co-equal owners; they are revocable, expiring capabilities constrained by a stateful policy monitor.

The policy monitor is not described as a vague “spending limit.” It must be *effect complete*: every reachable asset-moving side effect, including approvals, delegated calls, permit-style signatures, routers, and token-specific transfer semantics, must either be denied or charged against a typed loss budget. This yields a deliberately strong theorem: even if the session signature scheme is entirely broken, cumulative adversarial loss is bounded by the committed grant vector.

C. Contributions

This paper makes five contributions.

- C1. A new security problem and notion.** We formalize CDCD and define PQCC for assets moving between heterogeneous authorization domains.
- C2. A hybrid construction.** PREA composes exclusive UTXO–EVM transfer, a PQ-rooted smart account, irreversible authentication registration, and bounded classical delegation.
- C3. A compromise-resilient policy theorem.** We define BCD over an asset-loss vector and prove a cap theorem without assuming classical signature unforgeability.
- C4. Necessity and attack results.** We give minimal attacks on four tempting variants: classical destination ownership, mirrored balances, reversible PQ registration, and approval-permitting scalar caps.
- C5. An executable artifact and implementation roadmap.** A bounded checker produces concrete counterexample traces and validates the strict transition model; we map each invariant to consensus, EVM, wallet, and testing obligations in an existing Rust hybrid chain.

D. Scope and Venue Positioning

The contribution is a systems-security construction and formal security model, not a new lattice signature scheme. ML-DSA is treated as a standardized primitive under FIPS 204 [7]. The natural target is a security/systems venue after a complete prototype and evaluation. A cryptography venue would require a substantially deeper cryptographic construction or a full composable proof beyond the systems theorem presented here.

II. BACKGROUND AND RELATED WORK

A. Post-Quantum Signatures

Shor’s algorithm threatens discrete-logarithm signatures once a sufficiently capable quantum computer exists [8]. NIST standardized ML-DSA in FIPS 204 in 2024 [7]; the standard derives from the Dilithium design [9]. Our model uses ML-DSA as the root authorization primitive but does not claim that

every component of the EVM, network stack, or application is PQ secure.

B. Account Abstraction and Modular Accounts

ERC-4337 moves validation into smart accounts and an alternate UserOperation flow without consensus changes [1]. ERC-1271 standardizes contract-based signature validity [10]. ERC-6900 and ERC-7579 define modular account interfaces and motivate session keys, spending limits, and role-based controls [2], [11]. These mechanisms supply useful execution infrastructure, but they do not state how custody strength should compose with a source UTXO ledger or how loss should be bounded after assuming total compromise of the classical session scheme.

C. PQ Verification and Scheme Migration in EVM Systems

Poqeth evaluates PQ signature verification strategies in Ethereum and shows that verification cost is a central deployment concern [6]. EIP-7619 proposes a Falcon verification precompile [12]. Recent draft EIPs explore ECDSA-disabled delegation, native ML-DSA keys, scheme-agile transactions, and on-chain account configurations [3], [4], [5], [13]. These proposals materially narrow any novelty claim based solely on “PQ account abstraction” or “disable ECDSA after migration.” Our novelty claim is therefore narrower and compositional: preserving a PQ custody property across an exclusive UTXO–EVM value boundary while allowing a formally bounded classical capability.

D. Cross-Ledger and Cross-Domain Transfer

Atomic swaps and cross-chain communication research formalize fairness, liveness, and trust assumptions for moving value or messages across ledgers [14], [15]. PREA does not claim a trustless bridge between arbitrary independent chains. Its primary instantiation is a hybrid chain in which the UTXO and EVM transitions are committed by one consensus step. The question is orthogonal: after an atomic move succeeds, has the asset retained its intended authorization strength?

E. Comparison

III. SYSTEM AND ADVERSARY MODEL

A. Hybrid Ledger

The system exposes two authorization domains governed by a common finalized history.

- $\mathcal{L}_U$ is a transparent UTXO state machine. Standard spends require ML-DSA authorization.
- $\mathcal{L}_E$ is an EVM-compatible state machine with accounts, contracts, logs, and gas accounting.
- A composite transition T_i commits UTXO changes, deposit claims, EVM execution, withdrawals, and the corresponding roots atomically.

The construction also applies when the two domains are connected by a bridge with an authenticated finality oracle, but the proof then inherits the bridge’s trust and synchrony assumptions. To avoid hiding those assumptions, our main theorem parameterizes ledger failure by $\text{Adv}_{\text{ledger}}$.

TABLE I

POSITIONING AGAINST ADJACENT MECHANISMS. A CHECK MARK MEANS THE CITED MECHANISM DIRECTLY SPECIFIES THE PROPERTY, NOT MERELY THAT IT COULD BE EXTENDED TO APPROXIMATE IT. DRAFT EIPs ARE INCLUDED AS RELATED PROPOSALS RATHER THAN DEPLOYED STANDARDS.

Mechanism	Alternative account validation	PQ verification or key migration	Exclusive UTXO-EVM representation	Protocol-level closure of residual classical authority	Loss theorem under total classical compromise
ERC-4337 / modular accounts [1], [2]	✓	extensible	–	–	–
Poqeth / PQ precompile work [6], [12]	compatible	✓	–	–	–
EIP-7851 / EIP-8164 [3], [4]	✓	✓	–	✓	–
Scheme-agile / configured accounts [5], [13]	✓	extensible	–	partial	–
Atomic/cross-ledger transfer [14], [15]	–	–	✓	–	–
PREA (this work)	✓	ML-DSA root	✓	✓	✓

B. Asset Universe and Representation

Let $\mathcal{A}$ be a finite set of typed assets. Native units use an exact integer denomination. Fungible tokens are represented by (contract, amount), and non-fungible assets by (collection, tokenId). A state S contains four disjoint native-value compartments:

$$U(S), \quad L(S), \quad E(S), \quad B(S),$$

for spendable UTXOs, unclaimed deposit locks, EVM native balances, and burned value. For each issued native unit x , a representation function

$$\rho_S(x) \in \{U, L, E, B\}$$

must be single-valued.

For non-native EVM assets, custody is represented by account balances and ownership mappings. They are not mixed into the native issuance invariant.

C. Accounts and Authorization

A PQ account record is

$$R = (a, \text{ver}, \text{factory}, h_{\text{code}}, h_{\text{root}}, h_{\text{recovery}}, e, \text{frozen}),$$

where a is the deterministic account address and e is a monotonically increasing root epoch. The registry transition is one-way:

$$\text{Unregistered} \rightarrow \text{PqAccount}(R).$$

There is no transition back to a legacy ECDSA authorization type.

A root operation contains a domain-separated encoding of chain identity, genesis commitment, account, version, operation type, nonce, epoch, validity interval, call commitment, and policy identifier. It is accepted only if an ML-DSA signature verifies under the registered root.

D. Session Grants

A session grant g is

$$g = (\text{id}, pk_c, e, [t_0, t_1], \text{Cap}_g, \Gamma_g, n_g),$$

where pk_c is a classical key, e is the root epoch, $[t_0, t_1]$ is the validity interval, Cap_g is a typed budget vector, Γ_g is an

effect-complete policy, and n_g identifies an independent nonce space.

The cap is a vector rather than a single native-value number:

$$\text{Cap}_g = (c_{\text{native}}, \{c_{20,j}\}_j, \{c_{721,j}\}_j, \{c_{1155,j,k}\}_{j,k}, c_{\text{gas}}, c_{\text{calls}}).$$

ERC-721 components are sets or counters over explicitly authorized token identifiers; ERC-20 and ERC-1155 components are quantities.

E. Quantum-Classical Adversary

The adversary $\mathcal{A}$ is quantum polynomial time and has the following powers.

- 1) It forges arbitrary secp256k1 signatures and controls every registered classical session key.
- 2) It controls relayers, bundlers, RPC endpoints, transaction ordering within the ledger model, and any untrusted target contract.
- 3) It may submit arbitrary calldata and exploit every effect permitted by a policy.
- 4) It observes all public keys and prior signatures.

The adversary does not initially know the ML-DSA root secret. We assume ML-DSA unforgeability, collision resistance of domain-separated hashes, correct execution of the policy monitor, and finalized ledger safety. Failures of these assumptions appear explicitly in the theorem bound.

F. Why Assume Total Classical Break?

A session key is a compatibility feature, not a security foundation. Proving security under ordinary ECDSA unforgeability would only show that the system works before the quantum threat materializes. We instead grant the adversary a perfect classical signing oracle. The remaining guarantee is therefore a property of authorization architecture and stateful policy enforcement rather than of secp256k1.

IV. SECURITY DEFINITIONS

A. Exclusive Representation

Definition 1 (Exclusive representation). A native-asset execution satisfies ER if for every reachable finalized state S and every issued unit x , $\rho_S(x)$ is defined exactly once.

Equivalently, at aggregate precision,

$$I = U(S) + L(S) + E(S) + B(S), \quad (1)$$

where conversion between native UTXO units and EVM wei is exact and fixed.

B. Post-Quantum Custody Continuity

We define a game $\text{PQCC}_{\Pi}^A(\lambda)$ for construction Π .

- 1) The challenger generates an ML-DSA root and creates an asset under PQ UTXO custody.
- 2) $\mathcal{A}$ may schedule valid transfers between domains, request legitimate root-authorized setup operations, obtain classical grants chosen within the experiment, and forge all classical signatures.
- 3) The game exposes finalized state transitions and all public data.
- 4) $\mathcal{A}$ wins if it irreversibly acquires an asset or authority exceeding the authorized grant-loss vector without a valid root authorization.

Definition 2 (PQCC). Π provides post-quantum custody continuity if for all QPT adversaries,

$$\Pr[\text{PQCC}_{\Pi}^A(\lambda) = 1] \leq \text{negl}(\lambda) + \epsilon_{\text{ledger}} + \epsilon_{\text{policy}},$$

where explicitly granted loss is excluded from the win condition.

This is not a claim that the whole EVM is post-quantum. It is a custody statement about assets controlled by registered PQ accounts under a specified policy.

C. Bounded Classical Delegation

Let $\Delta(S_0, S_t, \mathcal{A})$ be the typed net asset loss from a protected account to adversary-controlled principals, including direct transfers, allowance-induced transfers, burns that benefit the adversary, and authority changes that enable later extraction. Let $\preceq$ be componentwise comparison over the loss vector.

Definition 3 (Effect completeness). A policy Γ_g is effect complete for grant g if every state transition reachable through an admitted operation is either (i) proven not to increase adversarial extraction capability, or (ii) charged to a corresponding component of the grant's cumulative budget before the external effect becomes reachable.

Definition 4 (BCD). A grant mechanism provides bounded classical delegation if, even with arbitrary valid classical signatures,

$$\Delta(S_0, S_t, \mathcal{A}) \preceq \sum_{g \in G_t} \text{Cap}_g$$

for every execution prefix, excluding root-authorized transfers and explicitly modeled gas expenditure.

The definition forces approvals and signature-based capabilities into the accounting model. A policy that only inspects top-level native value is not effect complete.

D. Authentication Monotonicity

Definition 5 (Authentication monotonicity). Once address a is finalized as a PQ account at epoch e , no later state admits a direct classical transaction or an account transition that restores unrestricted classical root authority for a .

This property overlaps recent ECDSA-deactivation proposals [3], [4]; in PREA it serves as one component of a broader cross-domain custody theorem.

V. THE PREA CONSTRUCTION

A. Deterministic Account and Registry

A fixed factory deploys account code with CREATE2. The salt binds chain genesis, the UTXO vault identity, account index, recovery commitment, and account version. A system registry records the expected factory and code hash, root-key commitment, recovery commitment, registration height, and frozen status. Registration is atomic with deployment and is authorized by an ML-DSA signature.

The registry is stored inside EVM state so it is covered by the existing state root, snapshot, reorganization, and archival machinery. It is not an off-chain list and does not introduce a second, loosely synchronized authentication database.

B. ML-DSA Verification Primitive

A fixed precompile validates a versioned input containing the expected root-key payload, a 64-byte domain-separated operation digest, the ML-DSA public key, and signature. The precompile reuses the native chain's deterministic portable verification path. Consensus and simulation register the same handler function.

Resource controls are consensus parameters: maximum verification calls per transaction and block, maximum authentication bytes, fixed malformed-input behavior, and an activation point. The construction does not assume verification is cheap; root operations are intentionally infrequent.

C. Root Operations

A root operation op commits to

domain, chainId, genesis, a , version, e ,
type, n , validAfter, validUntil, $H(\text{calls})$, $H(\text{policy})$.

The account checks epoch, nonce, validity, and hash commitments before invoking the verification precompile. Root-only operations include root rotation, recovery changes, session grant/revocation, account freeze/unfreeze, upgrade or migration, high-value asset transfer, and bridge withdrawal policy changes.

D. Direct Classical Authorization Closure

When transaction processing recovers a direct ECDSA sender a , the executor checks the in-execution registry view. If a is PQ registered, the transaction receives a deterministic invalid-transaction outcome rather than invalidating the containing block. The read occurs at one consensus-defined point for every verifier. Mempool admission may prefilter such transactions, but it is not a security boundary.

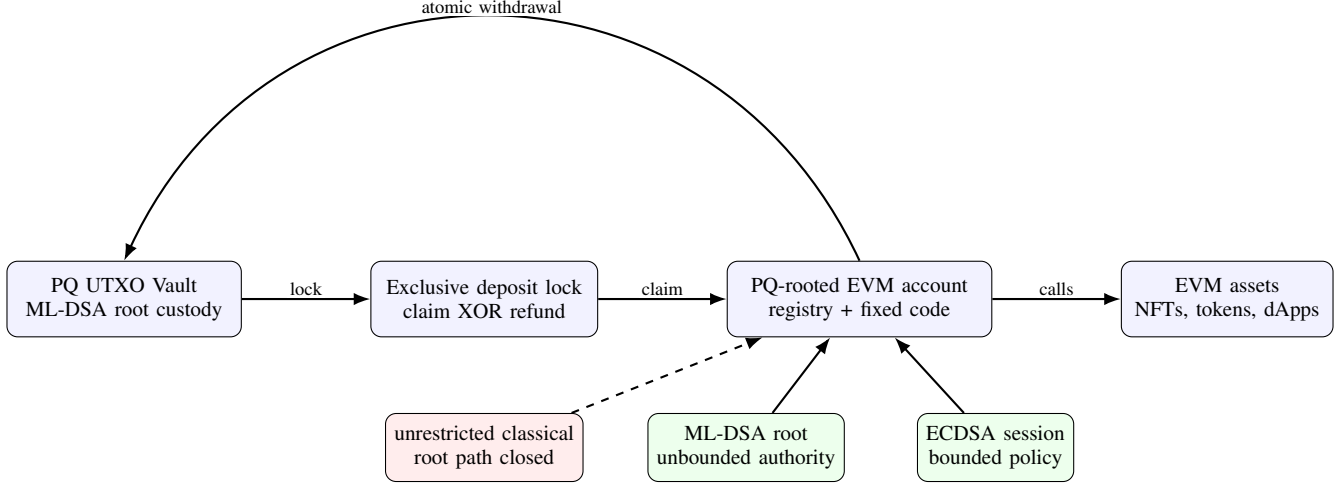


Fig. 1. PREA architecture. Native value is never mirrored. ML-DSA remains the root of authority after deposit; classical keys are explicit, bounded capabilities.

In a pure CREATE2 smart-account deployment, a matching direct ECDSA key is already computationally implausible. The protocol rule is defense in depth, prevents future authorization extensions from reopening the path, and makes authentication monotonicity explicit. It must not be misrepresented as the sole reason the account is PQ rooted.

E. Session Policy Monitor

A session operation carries a grant identifier, session nonce, validity information, typed call description, and classical signature. The account evaluates the operation in the following order:

- 1) load the grant under the current root epoch;
- 2) validate expiry and nonce;
- 3) decode the call using a grant-specific schema;
- 4) prove target, selector, recipient, asset, and amount authorization;
- 5) reserve cumulative budget in account state;
- 6) execute the call;
- 7) reconcile any operation-specific postcondition and finalize counters.

Unknown calldata is rejected. Generic delegatecall is rejected. Arbitrary routers, multicalls, and upgradeable proxies are denied unless a dedicated policy module fully decodes their internal effects and pins implementation identity. Unlimited approvals, `setApprovalForAll`, unconstrained Permit2 approvals, and opaque ERC-1271 hashes are denied by default.

F. ERC-1271 and Off-Chain Authority

A classical grant cannot label an arbitrary hash as a harmless login. For each supported purpose, the account reconstructs the expected typed-data hash from the full payload, domain, grant identifier, expiry, asset, recipient, and amount. It returns the ERC-1271 magic value only if that recomputed hash equals the verifier-provided hash and the purpose-specific cap is available. Unknown schemas fail closed.

G. Exclusive Deposit and Withdrawal

The deposit lock debits the UTXO owner and creates one unclaimed representation. Before timeout, exactly one valid claim may consume the lock and credit the PQ account; at or after timeout, exactly one refund may consume it and restore a UTXO. Withdrawal debits the EVM account and materializes a synthetic UTXO in the same composite transition. Claim/refund and debit/materialization are committed atomically.

A strict deployment binds the deposit lock to a finalized registry-record hash. Claim validation reads the selected parent's committed registry state, preventing nodes from consulting inconsistent or future EVM views.

VI. SECURITY ANALYSIS

We state the main results at a systems-cryptographic level. Full game definitions and proof details appear in the appendix. The proofs are modular: implementation bugs in policy decoding or ledger atomicity are not hidden inside a cryptographic assumption.

A. Supply and Exclusivity

Lemma 1 (Claim/refund exclusion). For any deposit lock with timeout τ , no accepting state can contain both a valid claim and a valid refund transition for that lock.

Proof sketch. Claim validity requires $d < \tau$ and refund validity requires $d \geq \tau$ for the consensus-defined DAA or height d . The predicates are disjoint. Both transitions consume the same unique outpoint, so duplicate application is also rejected. $\square$

Theorem 1 (Native supply conservation). Assume the genesis state satisfies Equation (1); all ordinary UTXO and EVM-native transfers conserve value modulo explicit fees/burns; claim and withdrawal are atomic; and conversion scale is exact. Then every reachable finalized state satisfies Equation (1) and exclusive representation.

Proof sketch. Induct over composite transitions. Ordinary moves preserve the sum. Lock creation decreases U and increases L by the same quantity. Claim decreases L and increases E . Refund decreases L and increases U . Withdrawal decreases E and increases U . Burn decreases its source and increases B . Atomicity excludes intermediate committed states, and unique outpoints exclude repeated consumption. $\square$

B. Root Authorization

Lemma 2 (Root-operation binding). If operation encoding is injective over its typed fields and the digest is collision resistant, a valid root signature for one chain, account, epoch, operation, nonce, call set, or policy cannot authorize a different tuple except with hash-collision probability.

Theorem 2 (PQ root unforgeability). An adversary that causes a non-root-authorized root operation to be accepted can be transformed into an adversary that either forges ML-DSA under the registered key, finds a collision in the operation commitment, violates registry/code integrity, or breaks ledger execution safety.

Proof sketch. The accepted operation must pass registry lookup, epoch/nonce checks, digest recomputation, and ML-DSA verification. If it differs from all signed operations, either the recomputed digest collides or the signature is a fresh forgery. If registry or code differs across honest verifiers, ledger/state integrity has failed. $\square$

C. Bounded Classical Delegation

Lemma 3 (Monotone budget accounting). For an effect-complete grant, every admitted adversarially beneficial effect increments a corresponding cumulative counter before control reaches an untrusted external target; counters never decrease within the grant epoch.

Theorem 3 (Classical compromise cap). Assume (i) the root is uncompromised, (ii) each active session policy is effect complete, (iii) external execution cannot re-enter before budget reservation or bypass account mediation, and (iv) grant counters are consensus state. Then, even if every classical signature is forgeable,

$$\Delta(S_0, S_t, \mathcal{A}) \preceq \sum_{g \in G_t} \text{Cap}_g.$$

Proof. Consider the sequence of accepted session operations. For the base state, loss and counters are zero. By effect completeness, an operation with no adversarially beneficial effect leaves the loss vector unchanged. Otherwise, the monitor reserves a componentwise budget increment δ_i before the effect, checks $C_g + \delta_i \preceq \text{Cap}_g$, and then permits at most the effects represented by δ_i . Reentrancy cannot observe unreserved capacity. Thus after each step, accumulated loss is bounded by accumulated counters, which are bounded by the cap. Summing disjoint grants yields the stated bound. The proof never invokes ECDSA unforgeability. $\square$

Corollary 1. If no classical grant is active, complete classical compromise alone yields zero authorized asset loss from the PQ account.

D. Authentication Monotonicity

Lemma 4. If registry records are immutable with respect to authorization type, account code cannot be replaced outside a root-authorized migration, and direct classical sends from registered principals are deterministically rejected, then a finalized PQ registration cannot transition back to unrestricted classical root authority.

E. Main Composition Theorem

Theorem 4 (Post-quantum custody continuity). Let Π_{PREA} instantiate: (1) exclusive representation and atomic cross-domain transitions; (2) ML-DSA-rooted account registration and root operations; (3) authentication monotonicity; and (4) effect-complete bounded classical delegation. Then for every QPT adversary $\mathcal{A}$,

$$\text{Adv}_{\Pi_{\text{PREA}}, \mathcal{A}}^{\text{PQCC}}(\lambda) \leq \epsilon_{\text{sig}} + \epsilon_{\text{hash}} + \epsilon_{\text{ledger}} + \epsilon_{\text{reg}} + \epsilon_{\text{pol}} + \text{negl}(\lambda), \quad (2)$$

where ϵ_{sig} is ML-DSA EUF-CMA advantage, ϵ_{hash} is the operation-hash collision advantage, and the remaining terms denote ledger, registry, and policy failures. Any value obtained through valid session grants is bounded by the aggregate grant caps and is not counted as a break.

Proof sketch. Replace successful unauthorized root actions by a root forgery/collision event using Theorem 2. Conditioned on no such event and on ledger/registry safety, exclusive representation prevents duplicate ownership and atomic transfer preserves the asset identity across domains. Authentication monotonicity prevents an unrestricted classical root path from reappearing. Theorem 3 bounds every operation reachable through forged classical credentials. Therefore an excess irreversible loss implies one of the enumerated bad events. $\square$

F. What the Theorem Does Not Prove

The theorem does not protect against an ML-DSA root compromise, malicious root-authorized grants, a vulnerable token contract that can arbitrarily rewrite balances, price volatility used to reinterpret a cap, or an incomplete policy module. It does not make ordinary EVM EOAs PQ secure and does not claim PQ-secure P2P transport or consensus.

VII. NECESSITY RESULTS AND ATTACKS

A. Unrestricted Classical Destination

Proposition 1. Any construction that credits a deposited asset to an account with an unrestricted classical authorization path cannot satisfy PQCC under the total-classical-break adversary.

Attack. Deposit one unit, forge the destination signature, and transfer it to the adversary. The bridge remains perfectly atomic and supply conserving, yet custody continuity fails.

B. Mirrored Balances

Proposition 2. A design that leaves a unit spendable in $\mathcal{L}_U$ while independently crediting it in $\mathcal{L}_E$ violates exclusive representation immediately.

This rules out “one balance shown in two ledgers” unless one side is a non-spendable view derived from the other.

TABLE II
BOUNDED-MODEL RESULTS. “FAIL” ROWS REPORT THE SHORTEST COUNTEREXAMPLE FOUND. THE MODEL IS EVIDENCE ABOUT THE ABSTRACT TRANSITION SYSTEM, NOT A PROOF OF THE RUST IMPLEMENTATION.

Variant	States	Edges	Result
PREA strict	201	688	Pass through depth 9
Classical destination	65	150	Fail, depth 4
Approval bypass	183	551	Fail, depth 6
Mirrored balance	14	14	Fail, depth 1
Auth downgrade	20	41	Fail, depth 2

C. Approval Bypass

Proposition 3. A scalar transfer cap does not imply bounded loss if the session may create an uncharged allowance, approval-for-all, delegatecall capability, or equivalent future extraction right.

Attack. A grant with cap one authorizes an unlimited approval without charging the cap. The approved spender later drains two or more units without another session operation. The account’s transfer counter remains within one while economic loss exceeds it.

D. Reversible Authentication

Proposition 4. If a PQ account may downgrade back to unrestricted ECDSA without a distinct, PQ-authorized migration to a new principal, then an attacker who can trigger or forge that downgrade defeats authentication monotonicity and subsequently drains the account.

E. Opaque Routers and Semantic Gaps

A top-level target/selector allowlist is insufficient when the target is a router, proxy, multicall, or signature gateway. The policy must reason about the actual asset effects or reject the operation. This is not merely a smart-contract coding recommendation; it is a precondition of Theorem 3.

VIII. EXECUTABLE BOUNDED MODEL

We provide a compact Python transition system as a supplementary artifact [16]. It models three native units, exclusive locks, claim/refund phases, PQ registration, root withdrawal, one-unit session grants, and an adversary with perfect classical forgery. Breadth-first search checks supply/exclusivity, authentication monotonicity, and bounded loss to depth nine.

The shortest traces are:

```
classical destination:
register -> lock(1) -> claim(1)
-> forged direct-ECDSA drain(1)

approval bypass:
register -> lock(2) -> claim(2)
-> grant(cap=1) -> approve(unlimited)
-> spender drain(2)

mirrored balance:
lock(1)

auth downgrade:
register -> downgrade-to-ECDSA
```

The strict model reaches 201 unique states and explores 688 edges without finding a violation. Because bounded exploration cannot establish unbounded correctness, the artifact is paired with the inductive proofs above and must be extended with implementation-level property tests.

IX. IMPLEMENTATION MAPPING

A. Current Baseline

The target system is a Rust chain derived from a GHOSTDAG-style UTXO core with an optional selected-parent EVM lane [17], [18]. The inspected design snapshot already contains:

- native ML-DSA transaction verification and known-answer tests;
- exclusive EVM deposit lock, claim/refund processing, and an EVM-to-UTXO withdrawal handler;
- a selected-parent EVM execution path and state commitment;
- the account, registry, and verification-precompile design documents.

The snapshot does *not* yet implement the PREA verification precompile, PQ registry, account contracts, direct-classical rejection, or the complete session policy. This paper is therefore an implementable design and formalization, not a claim that the production system already satisfies PQCC.

B. Consensus Components

The minimum implementation slice is:

- 1) a versioned ML-DSA verification precompile with deterministic gas and resource caps;
- 2) one shared registration function used by executor, `eth_call`, and gas estimation;
- 3) deterministic CREATE2 account/factory contracts with immutable code identity;
- 4) an EVM-state registry with one-way PQ registration;
- 5) per-transaction registry lookup and deterministic skipping of forbidden direct classical sends;
- 6) selected-parent-pinned registry validation for strict deposits;
- 7) stateful policy modules with typed counters and fail-closed decoders.

A forbidden direct-classical transaction must not invalidate an otherwise honest block. It follows the destination VM’s ordinary invalid-transaction/skip semantics so producers and verifiers agree.

C. Wallet and Relayer Components

The wallet maintains separate PQ root and EVM session profiles. It supports one-shot funding, session opening with explicit cap display, root-confirmed grant creation, and automatic withdrawal of excess native working balance. Session signatures never appear as unrestricted account owners. Relayers are untrusted and may be replaced without changing custody.

TABLE III

REQUIRED BENCHMARK MATRIX. VALUES ARE INTENTIONALLY ABSENT UNTIL MEASURED ON THE IMPLEMENTATION COMMIT USED FOR SUBMISSION.

Experiment	Baselines	Primary outputs
Root verification	native verifier, precompile	latency, gas, verifies/block
Root account call	EOA, ERC-4337 account	gas, bytes, end-to-end latency
Session transfer	EOA, modular account	gas, throughput, cap overhead
Deposit/withdraw	legacy bridge path	latency, reorg recovery, I/O
Policy fuzzing	unsafe variants	bypass count, coverage, traces
Soak/DoS	capped/uncapped builds	CPU, memory, block delay, rejection

D. Finality and Reorganization

Registration, claim, account state, receipt, withdrawal debit, and synthetic UTXO must revert as one canonical history changes. External irreversible actions—exchange credits, NFT-gated key release, physical delivery—use a finalized mapping, not mere payload inclusion or side-branch execution.

X. EVALUATION PLAN

A top-tier systems-security claim requires a complete implementation and measurement campaign. We separate measurements that can validate practicality from proofs that establish safety.

A. Research Questions

RQ1: Verification cost.

What are ML-DSA verification latency, gas calibration, authentication bytes, and block-level throughput across supported CPU architectures?

RQ2: Account overhead.

What additional gas, calldata, storage, and latency do root and session operations impose relative to EOA and ordinary smart-account baselines?

RQ3: Bridge continuity.

Under forced disconnects, reorgs, duplicate claims, and timeout races, does the implementation preserve the atomic invariants?

RQ4: Policy safety.

Do fuzzing and adversarial dApp suites find a path whose economic effect exceeds its typed budget?

RQ5: Availability.

How do PQ root bursts interact with payload byte caps, block gas, mempool policy, and public RPC resource limits?

B. Benchmark Matrix

Measurement pending: Populate Table III only after F003/F004, smart-account contracts, and the policy monitor are implemented at a pinned commit. Report medians, tail percentiles, hardware, compiler flags, dataset, and confidence intervals.

C. Correctness and Adversarial Tests

The release suite must include:

- FIPS 204 known-answer and malformed-input tests;
- differential tests between native and precompile verification;
- executor/simulation parity tests;
- stateful bridge fuzzing for conservation and claim/refund exclusion;
- session-policy fuzzing against approvals, Permit2, ERC-1271 confusion, multicall, delegatecall, proxies, reentrancy, and token callback hooks;
- cross-architecture deterministic execution tests;
- crash/restart, pruning, archive, snapshot, and reorg tests;
- independent smart-contract and consensus audits.

D. Artifact Requirements

The submission artifact should include source, reproducible build scripts, the bounded checker, property tests, benchmark harnesses, raw result files, plotting scripts, test vectors, container images, and a manifest binding every number in the paper to a command and commit.

XI. DISCUSSION

A. Why Not Remove ECDSA Globally?

Removing ECDSA or `ecrecover` from the entire EVM would break ordinary EOAs, relayers, token permits, and many existing contracts. PREA changes the authorization of registered protected accounts only. Unregistered EOAs and dApps remain in the classical compatibility domain. This isolates the PQ claim instead of overstating it.

B. Why Consensus Enforcement?

A smart account alone can use ML-DSA validation, but protocol-visible registration supplies three additional properties: a stable security label for wallets and bridges, an irreversible authentication state, and a consensus-defined rule that future transaction modes cannot silently restore unrestricted classical authority. The registry also lets a strict UTXO deposit commit to the exact destination security record.

C. Usability Tradeoff

Root signatures are large and expensive. The design intentionally moves high-frequency activity to session keys while quantifying the resulting exposure. Users can choose no session grant for cold custody, a narrow one-shot grant for minting, or a time-limited application grant. The UI must display both the cap and whether a policy includes transferable approvals or opaque targets.

D. Asset Valuation

A token-unit cap is precise but does not bound fiat value under changing prices. A deployment may use conservative oracle-denominated caps, but then oracle integrity enters the theorem. Our core definition therefore uses asset units and NFT identities; any scalar economic valuation is an explicit extension.

E. Liveness

The safety proof tolerates unavailable relayers and offline roots, but liveness does not. Recovery, bundler diversity, root-operation fee funding, and PQ verification capacity require separate engineering. A fail-closed policy may reject unfamiliar dApps until a reviewed module is installed; this is a deliberate security/compatibility tradeoff.

F. CRYPTO-Level Extension Path

For a cryptography-focused contribution, one could seek a new primitive that succinctly proves a stateful policy-compliant ML-DSA authorization, aggregates many root operations, or offers composable cross-domain custody under a formal ledger functionality. Those are distinct research problems. The present work’s central novelty is the systems-security composition and bounded-loss theorem.

XII. LIMITATIONS AND RESPONSIBLE CLAIMS

The following claims would be incorrect:

- “The EVM is fully post-quantum.” Ordinary EOAs and many application signatures remain classical.
- “A PQ precompile alone preserves custody.” Destination authority and policy semantics are equally necessary.
- “Any selector allowlist bounds damage.” Opaque routers, approvals, and callbacks can bypass superficial checks.
- “Bounded model checking proves the implementation.” It does not.
- “The paper proves ledger finality.” It parameterizes finality failure.

The paper’s defensible claim is narrower: for assets owned by a registered PREA account, under the stated ledger, primitive, registry, and policy assumptions, classical-key compromise cannot cause loss beyond explicit grant caps.

XIII. CONCLUSION

Hybrid chains create a security boundary that signature-migration discussions often miss. A value transfer can be atomic, supply conserving, and still downgrade custody from a PQ authorization domain to a forgeable classical account. We formalized this problem as post-quantum custody continuity and presented PREA, a construction that preserves exclusive value representation, roots destination authority in ML-DSA, closes residual unrestricted classical paths, and retains practical EVM compatibility through bounded classical delegation.

The strongest result is intentionally independent of ECDSA security: if the session-policy monitor is effect complete, total classical compromise cannot exceed a typed, stateful grant cap. The corresponding necessity results show why

mirrored balances, unrestricted destination EOAs, reversible authentication, and approval-permitting scalar caps fail. A bounded artifact already generates these counterexamples; the remaining submission gate is a complete implementation with reproducible performance and adversarial evaluation.

APPENDIX A

FORMAL GAMES AND PROOF DETAILS

A. Ledger Functionality

We model a finalized composite ledger as a deterministic transition function

$$\mathcal{F}_{\text{HE}} : (S, T) \mapsto S' \text{ or } \perp,$$

where $S = (S_U, S_E, S_R)$ contains UTXO, EVM, and registry state. A transaction batch T is canonically ordered by the underlying consensus. The functionality exposes a finalization predicate $\text{Finalize}(i)$ and guarantees prefix agreement except with probability ϵ_{ledger} .

A deposit identifier d has a state in

$$\{\text{Absent}, \text{Locked}, \text{Claimed}, \text{Refunded}\}.$$

Allowed transitions are

$$\text{Absent} \rightarrow \text{Locked} \rightarrow \begin{cases} \text{Claimed}, & h < \tau_d, \\ \text{Refunded}, & h \geq \tau_d. \end{cases}$$

No outgoing transition exists from Claimed or Refunded.

B. Root Authorization Game

The challenger generates $(pk, sk) \leftarrow \text{KeyGen}(1^\lambda)$ and installs $H(pk)$ in the registry. The adversary obtains signatures on chosen valid operation encodings. It wins by causing acceptance of an operation encoding not previously signed under the current root epoch. The reduction to ML-DSA EUF-CMA is standard after conditioning on digest injectivity/collision resistance and correct state lookup.

C. PQCC Experiment

The experiment tracks a protected asset set P and grant set G . It grants the adversary:

- all public chain data and state queries;
- a classical signing oracle that returns arbitrary valid signatures;
- control of relayer and transaction scheduling interfaces;
- a root-operation oracle restricted to operations explicitly approved by the experiment, such as creating a grant with a known cap;
- chosen dApp contracts within the policy’s target universe.

The win predicate is true if a finalized state gives the adversary a protected asset, an unlimited extraction authority, a root/upgrade/recovery authority, or a bridge withdrawal right not derivable from an experiment-approved root operation and not covered by the cumulative grant vector.

D. Hybrid Argument for Theorem 4

Game 0.

The real PQCC experiment.

Game 1.

Abort on a finalized ledger safety/atomicity failure. Difference is at most $\text{Adv}_{\text{ledger}}$.

Game 2.

Abort on inconsistent or mutable registry/code state. Difference is at most $\text{Adv}_{\text{registry}}$.

Game 3.

Abort on a fresh accepted root operation. By Theorem 2, the difference is bounded by ML-DSA forgery plus hash collision.

Game 4.

Replace every classical authorization by adversarial choice, which does not change the experiment because classical forgery was already granted. Apply Theorem 3 to bound all session-mediated loss.

In Game 4, exclusive representation and authentication monotonicity leave no other acquisition path. Any remaining win is a policy-completeness violation, contributing $\text{Adv}_{\text{policy}}$.

APPENDIX B NORMATIVE POLICY CONDITIONS

A policy module used in the theorem must satisfy all of the following.

- 1) **Total decoding:** accepted calldata has a unique schema interpretation; malformed or trailing data is rejected.
- 2) **Target identity:** target address and, where relevant, implementation code hash are pinned.
- 3) **Recipient binding:** every asset recipient is either fixed, proven in a committed set, or the protected account itself.
- 4) **Typed accounting:** native, ERC-20, ERC-721, and ERC-1155 effects update distinct counters.
- 5) **Capability accounting:** approvals, permits, operators, and signatures that create future extraction rights are treated as immediate worst-case loss or denied.
- 6) **No generic delegatecall:** execution cannot borrow unreviewed code in the account's storage context.
- 7) **Reentrancy safety:** budget is reserved before any external call and cannot be rolled back independently of the call's effect.
- 8) **Epoch and nonce binding:** a grant cannot survive root rotation unless explicitly renewed.
- 9) **Fail-closed evolution:** unknown token hooks, proxy upgrades, schemas, or router versions are rejected.

A practical implementation should provide small audited modules instead of a universal calldata interpreter.

APPENDIX C EXECUTABLE MODEL REPRODUCTION

Run:

```
python3 model_check.py
```

Expected summary:

```
PREA-strict: PASS through depth 9
(201 states, 688 edges)
naive-classical-destination: FAIL depth 4
naive-session-approval: FAIL depth 6
naive-mirrored-balance: FAIL depth 1
naive-auth-downgrade: FAIL depth 2
```

The output file `model_results.json` contains full states and shortest traces. Increasing asset units and depth is useful for regression, but exhaustive state growth is exponential; implementation verification should combine property-based testing, symbolic execution of policy modules, and invariants in the EVM/UTXO transition code.

APPENDIX D SUBMISSION COMPLETION CHECKLIST

Before claiming a completed IEEE S&P or ACM CCS evaluation, the authors should provide:

- 1) a pinned public implementation of the precompile, registry, accounts, and policies;
- 2) a machine-checked or mechanized refinement of at least the core state transition invariants;
- 3) measured root/session costs on multiple CPUs and realistic block limits;
- 4) differential execution tests across consensus and simulation;
- 5) adversarial policy tests including approval and router bypasses;
- 6) reorganization and crash-consistency experiments;
- 7) independent security review and disclosure of unresolved findings;
- 8) a reproducible artifact evaluated from a clean environment;
- 9) a tightened related-work review immediately before submission because PQ-Ethereum proposals are evolving rapidly.

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